

The steeper 1/f slope of MEG power spectrum is associated with stronger gamma synchrony, measured via inter-trial phase coherence (ITPC) in the 40-Hz auditory steady-state response

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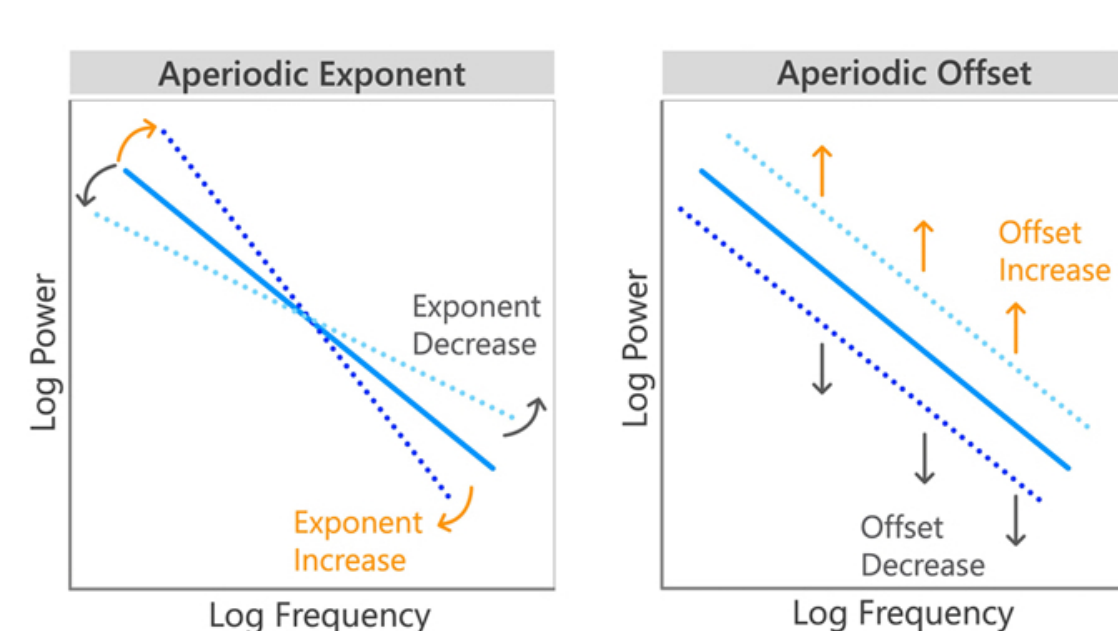
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Hypothesis. We explored whether a steeper slope of the 1/f PSD component (i.e., a larger spectral exponent) in MEG signals is associated with higher auditory steady-state response (ASSR) synchrony, as measured by inter-trial phase coherence (ITPC) during 40 Hz auditory stimulation.

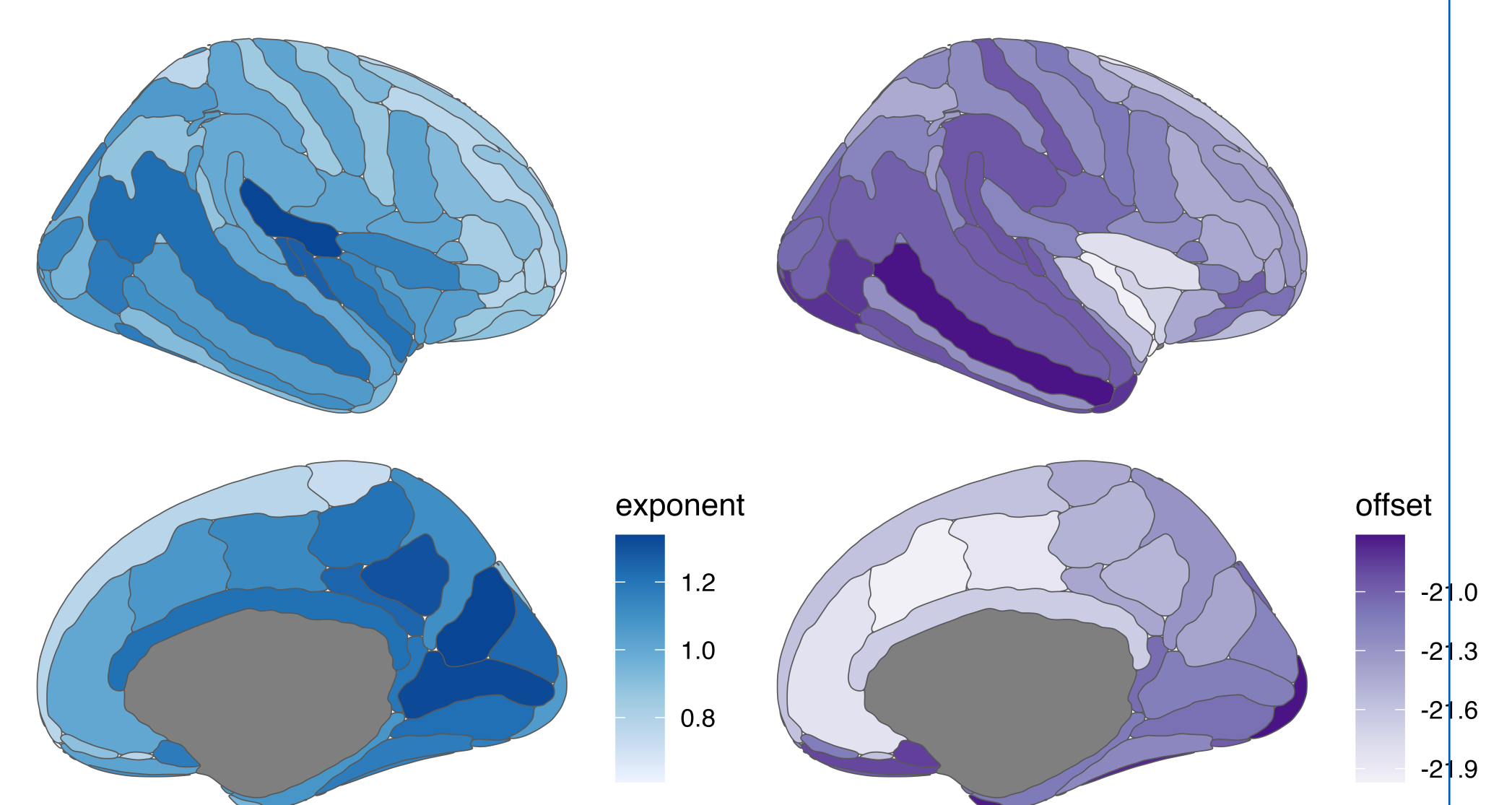
Introduction

- Power spectral density (PSD) of electrophysiological activity may be parametrized by oscillatory components and aperiodic component, where the power of the signal decays with a 1/f power law
- The timescale and magnitude of this aperiodic activity can be described by the slope and offset of this 1/f decay [2]

- Simplified formula describing 1/f component of the PSD of the signal: $AP(f) = b - \log(k + f^\chi)$; The exponent χ is related to the negative slope of the power spectrum in log-log space, that is, the inverse relationship between power and frequency; The offset b describes the uniform shift in power across frequencies
- These parameters are most likely governed by waxing and waning of oscillatory components, and can be associated to biological mechanisms (e.g. excitation:inhibition balance as summed by [7], cognitive processes, and group differences).
- Graphical relationship of spectral power and offset and exponent (from [3])



1 Spectral Parametrization. Mean values of estimated exponent and offset describing the 1/f part of spectral power across different regions (right hemisphere).



- Synchronization of brain activity in the gamma frequency is a hallmark of brain function, and an indicator of malfunctioning [5].
- Auditory stimulation in the gamma range is a reliable way to experimentally investigate this phenomenon, looking at auditory steady-state responses (ASSR), the effect of entrainment of brain activity to the frequency and phase of external auditory stimuli.
- ASSR are usually investigated by looking at the inter-trial phase coherence (ITPC) [5].

Data

- We re-analysed MEG data from n=15 healthy participants, stimulated with a sound wave modulated at 40 Hz, presented in 180 trials for one second, with the onset of two consecutive trials separated by at least 2 s [5].
- The data were preprocessed, projected into Destrieux brain atlas

Conclusion

- The differences observed might simply reflect increased "noise" in the time-domain signal, potentially diminishing SNR and synchrony. Alternatively, from a biological perspective, this effect might be linked with excitation-inhibition balance, which influence both synchrony measures and spectral slope.
- The effects of the aperiodic slope on the estimation of the instantaneous phase and the ITPC [6] could be considered in further analysis as well as a effect of higher gamma power for aperiodic slope estimation.

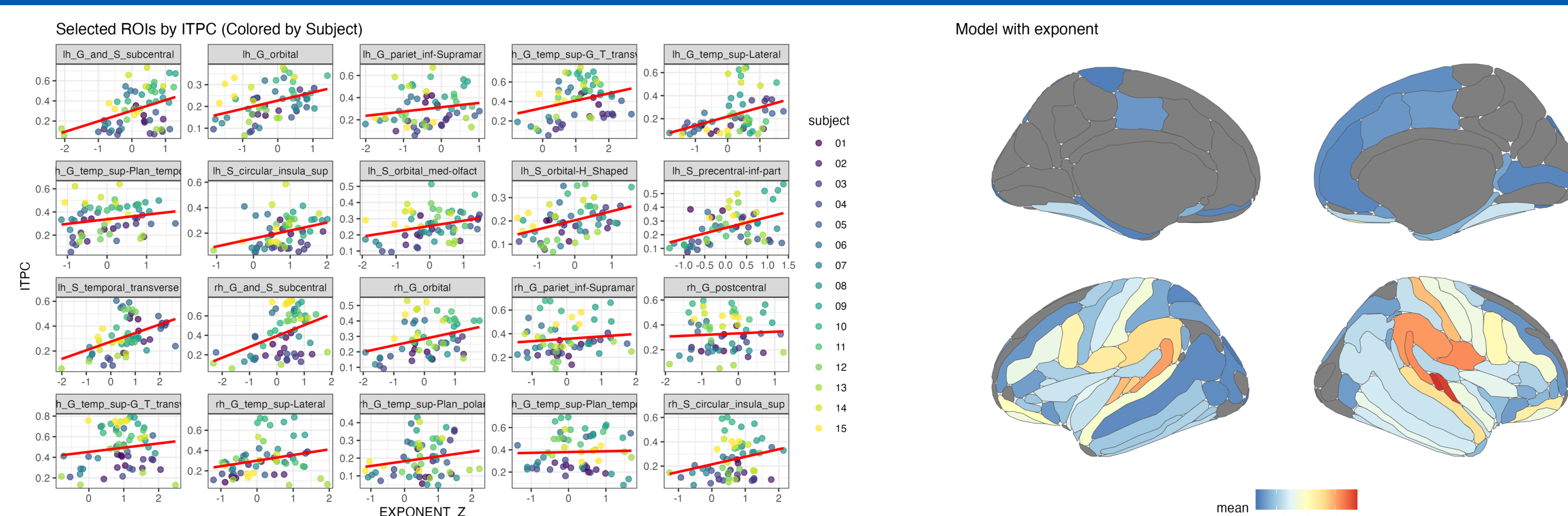
Acknowledgments: For visualising the results ggseg library was used [4]

References

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Relationship between the slope (parametrized by exponent and the estimated ITPC). Figure

2 on the left: Relationship between Spectral Exponent (z-scored) and ITPC across selected ROIs. Red line indicates linear fit. Data points colored by subject.



Methods

- The PSD of the spectrum were computed for 1s second pre-stimulus period and stimulation period for each trial (and averaged later) and for each region. The slope and exponent were computed using specparam toolbox [2].
- ITPC was computed, the value averaged in the intervals $[-500, -200]$ ms for prestimuli and $[400, 700]$ ms after sound onset for each of the regions using the wavelet approach described in [5].
- The influence of the stimulus and the exponent on the ITPC were tested by modelling the different relationships with an integrative Bayesian Multilevel Model [1]
- This methods allows for better integration of the spatial information and control of multiplicity avoiding information loss of univariate approach followed by the correction for multiple testing.
- Effect estimation and region-level inferences are based on posterior distributions of group differences.

Following models were tested:

- ITPC $\sim$ S (Stimulus)
- ITPC $\sim$ Exponent + Stimulus
- Exponent $\sim$ Stimulus
- Offset $\sim$ Stimulus

- For investigation of the the relationship we fit the model

$$ITPC_{stim} \sim 1 + exponent_z + (1 + exponent_z | subject) + (1 | ROI)$$

with using z-scored Exponent. The model was fitted using the brms package and [1].

Results (Box 2)

- Exponent and the offset of the spectrum are different across the brain (Box 1);
- Strong effect of the stimulus (S) on the ITPC (contrast baseline-stimulus), in areas like the auditory regions, but also in temporal lobe, inferior part of primary sensory-motor regions and inferior frontal regions; negative values indicate higher ITPC during stimulation.
- No particular evidence was found for the model $Exponent \sim Stimulus$, namely the stimulation did not influence the exponent per se.
- No significant evidence was found for any of the models involving the Offset either as a predicted variable, and as a predictor, suggesting a marginal role for it in the auditory entrainment.
- Looking for a raw relationship between exponent and ITPC shows a positive relationship between the Exponent and the ITPC, suggesting that higher Exponent is associated with higher ITPC, i.e. higher synchrony during stimulation.
- Hierarchical Bayesian modelling allows to estimate the posterior distribution of the coefficients, which can be used to assess the significance of the effect. It also uses pooling of the data across subjects and regions. The posterior distribution of the coefficient is shown in the Box 2, on the right.